



New Trends in BIS

Memory Issues In Business Information Systems

Prepared by:

Karim Abdelhamid Mahmoud Kamel Deheya
221004084

Kareem Mohamed El Shafei 221005696

Seif Amr Attia 221005621

Course Instructor:

Dr. Ashraf Abdulaliem

Teaching Assistant:

TA Angie Abouzaid

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Introduction

Background

Memory issues are not only related to human thinking, but they also play a very important role in Business Information Systems (BIS). In organizations, memory can be understood as the ability of systems to store, manage, and retrieve data efficiently. Any problem in this process can lead to serious issues such as slow performance, data loss, or incorrect decision-making.

In modern businesses, companies depend heavily on information systems such as databases, cloud storage, and enterprise systems to manage their operations. These systems are responsible for handling large amounts of data every day. However, when there are memory issues, such as insufficient memory, poor data management, or system overload, the performance of these systems becomes weak. For example, a system may become slow when retrieving data, or it may fail to process large datasets efficiently.

There are several reasons behind memory issues in BIS. One major factor is the rapid growth of data. Organizations today deal with huge volumes of information, and if the system is not designed to handle this data properly, it may face memory limitations. Another reason is poor system design or lack of optimization, where memory resources are not used efficiently. In addition, outdated hardware or insufficient RAM can also cause serious performance problems.

Purpose of the research

The purpose of this research is to explore the different types of memory issues in Business Information Systems and understand how they affect system performance and business operations. It also aims to highlight simple and effective solutions, such as improving system design, upgrading hardware, and using modern technologies like cloud computing to manage memory more efficiently.

Significance

Memory issues in Business Information Systems are very important because they directly affect how organizations operate. When systems suffer from memory problems, it can lead to slow performance, delays in processing data, and even system crashes. This can negatively impact productivity and reduce the efficiency of business operations.

History

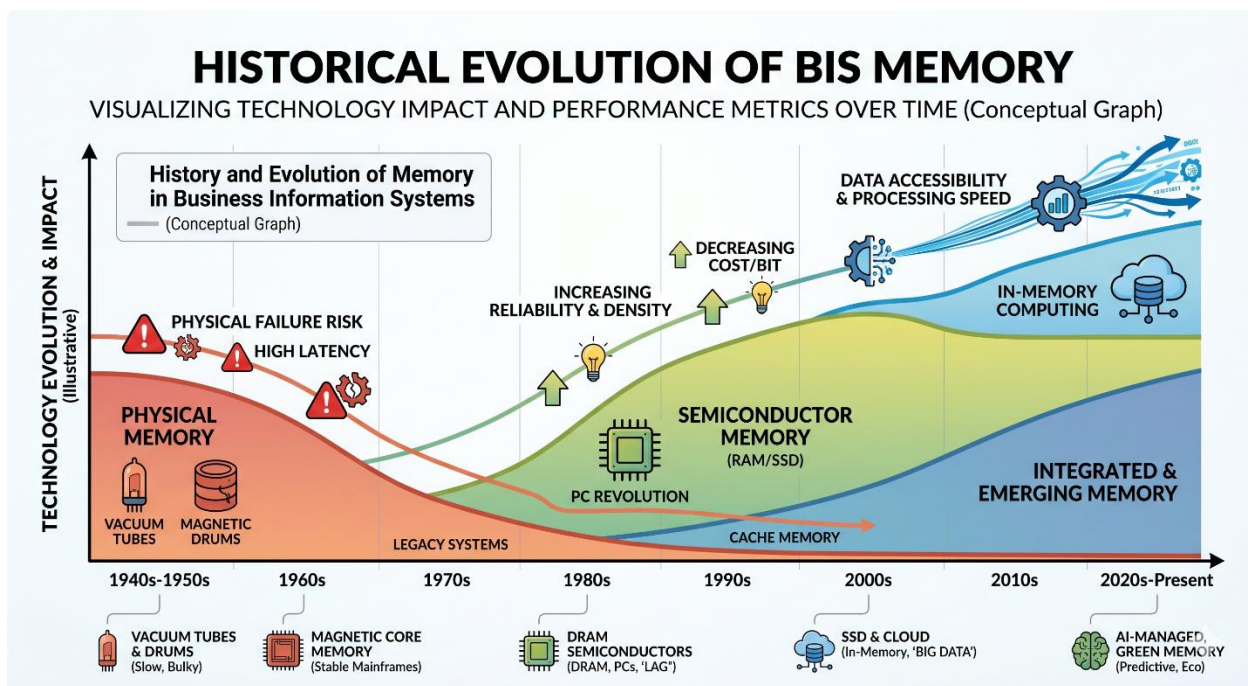
The journey of memory in Business Information Systems (BIS) is a fascinating transformation from massive, mechanical hardware to invisible, lightning-fast cloud structures. In the 1940s and 1950s, the very first business computers, such as the UNIVAC, relied on vacuum tubes and magnetic drums. These components were incredibly bulky, generated immense heat, and had very little storage capacity. During this era, "memory" was a physical struggle; a single machine could occupy an entire room just to process basic arithmetic for census data or simple accounting. The lack of reliable memory meant that businesses could only automate the most basic tasks, and the risk of hardware failure was a constant threat to data integrity.

By the 1960s and 1970s, a major breakthrough occurred with the invention of magnetic core memory. This technology used tiny magnetic rings to store bits of data. It was significantly more reliable and faster than vacuum tubes, allowing businesses to move toward "real-time" processing. For the first time, large corporations could manage complex payroll systems and inventory tracking across multiple departments. This era marked the birth of the modern mainframe, where memory became stable enough to support the first generation of database management systems. Companies began to realize that digital memory was not just a technical requirement but a strategic asset that could speed up decision-making.

The 1980s and 1990s brought the "Silicon Revolution," introducing Dynamic Random Access Memory (DRAM). As microchips became smaller and cheaper, the power of a mainframe was condensed into a desktop PC. This shift democratized business information systems; small businesses could now afford the memory capacity needed to run spreadsheets and local databases. However, this period also introduced new challenges. As software grew more complex, "memory

bloat" became a common issue. Operating systems like Windows and early ERP software demanded more RAM than most computers had, leading to the first widespread observations of system "lag" and the need for virtual memory management.

In the modern era, from the 2000s to today, we have moved into the age of Flash memory, Solid State Drives (SSDs), and Cloud-based memory. Today's Business Information Systems no longer wait for a spinning disk to find data. Instead, technologies like "In-Memory Computing" allow entire global databases to sit directly in the system's active memory for instant access. We are now seeing the integration of Artificial Intelligence, where memory systems are "smart" enough to predict which data a business user will need before they even click a button. From the glowing vacuum tubes of the 50s to the invisible servers of the cloud, memory has evolved from a physical limitation into the digital heartbeat of global commerce.



Theoretical Framework

Concept

- **Memory Allocation:** This is how the system "gives" pieces of memory to different business tasks (like opening a spreadsheet or processing a sale).
- **Memory Paging:** This is a technique where the system breaks data into small, equal-sized "pages." It allows a business system to store data in different physical locations of the RAM while making it look like one continuous block to the software.
- **Garbage Collection:** An automatic process in modern business software (like Java-based ERPs) that looks for memory that is no longer being used and "reclaims" it so other tasks can use it.
- **Memory Swapping:** When the RAM is full, the system "swaps" inactive business tasks (like a background report you aren't looking at) onto the hard drive to make room for the task you are currently using.
- **Concurrency Control:** In systems like banking, this ensures that when two people try to access the same piece of memory at once (like two people buying the last item in stock), the memory is "locked" correctly so the data doesn't get corrupted

Theories

1 - The Von Neumann Bottleneck (The Efficiency Theory)

Founder: John von Neumann (Architecture, 1945) and John Backus (who coined the "Bottleneck" term in 1977).

The Date: 1945 / 1977.

The Concept: This theory explains that a system slows down because the "brain" (CPU) can process data much faster than the "hallway" (the Bus) can send it from the memory.

Implementation: Every modern PC and smartphone is built on this architecture. To "fix" it, engineers implemented CPU Caching (L1, L2, L3 cache) to keep data closer to the brain.

2 - Organizational Memory (The Management Theory)

Founders: James P. Walsh and Gerardo Rivera Ungson.

The Date: January 1991.

The Concept: This suggests that a business's information system isn't just hardware; it is a "digital brain" that stores the company's past experiences and knowledge to help make future decisions.

Implementation: This is the foundation of Knowledge Management Systems (KMS). NASA used this to preserve engineering knowledge from the Apollo era, and Toyota uses it to ensure old mistakes are never repeated.

3 - The Locality of Reference (The Speed Theory)

Founder: Peter J. Denning.

The Date: 1967.

The Concept: This theory states that if a computer uses a piece of data, it is very likely to need that same data (or data saved right next to it) again very soon.

Implementation: This is why Web Browsers (like Chrome) use "Cache." It saves the website logo on your computer so it doesn't have to download it every time you click a new page.

4 - The Memory Hierarchy Theory

Founder: Various researchers (became standard in the 1960s with the IBM System/360).

The Date: Developed throughout the 1960s.

The Concept: Computers should use different types of memory: small/fast/expensive memory for active work and large/slow/cheap memory for storage.

Implementation: Every business computer uses this by having RAM for active apps and Hard Drives for saved files.

5 - Amdahl's Law

- **Founder:** Gene Amdahl.
- **The Date:** 1967.
- **The Concept:** Improving one part of a system (like memory speed) only helps the total speed up to a certain point if the rest of the system stays slow.
- **Implementation:** Businesses use this to decide if it's worth spending money on more RAM. If the internet connection is slow, adding more memory won't make the business faster.

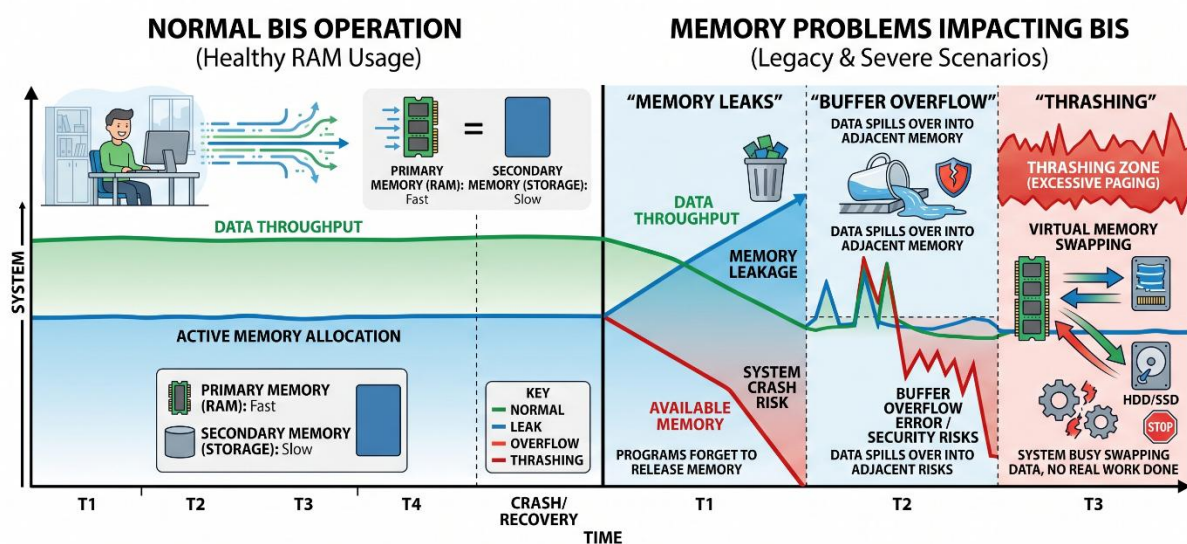
Features of Memory Issues

- **Latency (Lag):** A delay between clicking a button and the system responding.
- **Volatility:** Memory is "temporary." If the power goes out and the system hasn't saved data to a hard drive, that information is lost forever.
- **Fragmentation:** Over time, memory gets "messy" with small gaps of empty space. This makes it harder for the system to find enough room for large new tasks
- **Inelasticity (Scalability Failure):** This occurs when a system cannot "stretch" to handle sudden spikes in traffic, such as during a Black Friday sale. Instead of expanding, the memory remains rigid, causing the system to reject new customers or freeze entirely.
- **Persistence Delay:** This is a failure in the transition between "working memory" and "storage memory." If the system cannot move data to permanent storage fast enough during a crash, critical business information is permanently lost.
- **Corruption (Data Integrity Failure):** This happens when memory lacks "Error Correction Code" (ECC) or self-healing capabilities. A single "flipped bit" can silently change important numbers, such as turning a \$1,000 invoice into a \$1,000,000 error.

Types of Memory Problems

- **Memory Leaks:** This happens when a program asks for memory but forgets to give it back when it's finished. Eventually, the system runs out of memory and crashes.
- **Buffer Overflow:** This is like trying to pour a gallon of water into a tiny cup. Data spills over into other parts of the memory, which can cause security holes or "blue screens."
- **Thrashing:** This occurs when the system is so busy moving data in and out of memory that it stops doing any real work.
- **Memory Fragmentation:** This occurs when memory is broken into many small, non-continuous pieces. Even if you have "free space," the system might crash because it can't find a single continuous block large enough to run a big business application.
- **Segmentation Fault:** This happens when a business application tries to access a part of the memory that it is not allowed to touch (like a spreadsheet trying to read a password stored by the operating system). The system usually shuts the program down immediately to protect data security.

VISUALIZING MEMORY PROBLEMS IN BIS



Current Trends

- **AI-Optimized Memory:** New systems use Artificial Intelligence to predict which data a manager will need next, moving it into "fast memory" before they even ask for it.
- **High-Bandwidth Memory (HBM3):** Specialized memory designed for heavy AI tasks and real-time data analytics.
- **Sustainable "Green" Memory:** Companies are switching to memory chips that use much less electricity to help meet environmental goals.
- **Non-Volatile RAM (NVRAM):** This is a "dream" technology for businesses. It is as fast as RAM but doesn't lose data when the power goes out. It aims to eliminate the "saving" process entirely.
- **Memory-Centric Architecture:** Traditionally, the CPU was the center of the computer. New trends put the **memory** at the center, allowing multiple processors to access a massive "pool" of shared data simultaneously.
- **Edge Memory:** With the rise of IoT (Internet of Things), businesses are moving memory to the "edge"—putting data storage closer to the physical location (like a factory floor or a shipping port) to reduce the time it takes for data to travel.

Case Studies

Example 1: SAP (In-Memory Computing)

The Problem: Traditional databases were too slow for global companies to see their sales in real-time. The Solution: SAP created HANA, a system that keeps all business data in the "active memory" (RAM) instead of on slow hard drives. The Result: Companies like Walmart can now analyze millions of transactions in seconds rather than hours, allowing them to adjust prices or stock instantly.

Example 2: Netflix (Memory Caching)

The Problem: With millions of people watching videos, the system's memory could easily get overwhelmed, causing movies to buffer or fail. The Solution: Netflix uses a technique called EVCache. They store "popular" data in a special layer of high-speed memory. The Result: By managing their memory this way, they reduced the load on their main databases and ensured that videos start almost instantly for users worldwide.

Example 3: Creative Cloud Performance

The Challenge: When Adobe moved their software to the cloud, users complained that editing large video files was too slow because the "memory" was located in distant servers. **The Solution:** Adobe implemented **Intelligent Caching**. The system uses the user's local device memory to store the parts of the file they are currently working on, while the "heavy" data remains in the cloud. **The Result:** This hybrid memory approach allowed for a smooth user experience, proving that even cloud-based businesses must carefully manage local memory issues to keep customers happy.

Advantages and disadvantages

Memory Type	Advantages	Disadvantages
Cache Memory	Extremely fast; provides nearly instant access to common business commands.	Very expensive; has very limited storage capacity.
Primary Memory (RAM)	Allows for smooth multitasking; fast enough for real-time data entry.	Volatile (data is lost if power fails); more expensive than long-term storage.
Secondary Memory (SSD/HDD)	Permanent storage; can hold massive amounts of company records (Big Data) cheaply.	Much slower than RAM; mechanical parts (in HDDs) can wear out over time.
Virtual Memory	Allows computers to run large programs that are bigger than the actual RAM installed.	Significantly slows down the system if used too much; can lead to "thrashing."

Conclusion

In conclusion, managing memory is a fundamental requirement for the success of any Business Information System (BIS). As explored in this research, memory has evolved from the slow, physical vacuum tubes and magnetic drums of the 1940s into the lightning-fast, AI-driven cloud architectures of today. Theoretical frameworks like the **Von Neumann Bottleneck** and **Organizational Memory** demonstrate that memory is not just a hardware specification; it is a critical bottleneck for efficiency and a digital brain for corporate knowledge.

While modern features like **Scalability** and **In-Memory Computing** offer immense power, they also introduce risks such as **Memory Leaks** and **Fragmentation**, which can lead to system crashes or severe security vulnerabilities. The success stories of companies like **Walmart** and **Netflix** prove that optimizing memory leads to faster data processing and better customer experiences, while historical failures, such as the **32-bit banking crash**, remind us of the high cost of poor memory planning.

As we look toward 2026, trends like **Sustainable Green Memory** and **AI-Optimized Paths** suggest that the future of BIS will focus on making memory not only faster but also smarter and more energy-efficient. For any business to remain competitive, it must treat its system

memory as a strategic asset balancing the speed of **Cache** and **RAM** with the reliability of **Secondary Storage** to ensure that vital information is always available exactly when it is needed

References

- **Laudon, K. C., & Laudon, J. P. (2024).** *Management Information Systems: Managing the Digital Firm* (18th ed.). Pearson.
- **Stallings, W. (2022).** *Computer Organization and Architecture* (11th ed.). Pearson.
- **Tanenbaum, A. S., & Bos, H. (2023).** *Modern Operating Systems* (5th ed.). Pearson.
- **SAP SE. (2025).** *SAP HANA: The Evolution of In-Memory Computing for Modern Business*. Official Whitepaper.
- **Netflix Technology Blog. (2021).** *Evolution of Edge Caching at Netflix.* > An article explaining how **Netflix** uses **EVCache** and high-speed memory to prevent buffering and system crashes during peak hours.
- **Amazon Web Services (AWS). (2026).** *High-Bandwidth Memory (HBM3) and the Future of Cloud Computing.* > Provides insights into the current trends of AI-optimized memory and "Green" sustainable tech.
- **Walsh, J. P., & Ungson, G. R. (1991).** "Organizational Memory." *Academy of Management Review*.